Feng Ruiqi

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EDUCATION

National University of Singapore Aug. 2026 – Jun. 2027
Master of Computing in Computer Science

Huazhong University of Science and Technology Sept. 2022 – Jun. 2026
B.Eng. in Computer Science and Technology GPA: 4.08/5.0 (86/100)

- First Prize, National English Competition for College Students; University Scholarship.
- Selected for the 2026 Double First-Class Undergraduate Top Innovative Talent Program, with funding for a one-year exchange at NUS.

National University of Singapore, School of Computing Aug. 2025 – May 2026
One-Year Exchange Program GPA: 4.67/5.0

EXPERIENCE

Xinda Company, Changjiang Design Group *AI Agent R&D Intern* Jun. 2026 – Aug. 2026

Project: A LangGraph-powered DeerFlow flood-control agent platform integrating LLMs, multi-Skill orchestration, and hydrological, meteorological, and engineering data services for monitoring, analysis, forecasting, alerts, reservoir scheduling, and decision support.

- **Agent Evaluation System:** Built an automated evaluation system for 10 categories of flood-control tasks with Trace replay; designed outcome and process metrics for task completion, format compliance, planning efficiency, and Skill precision, supporting Excel import, filtering, concurrent execution, checkpoint recovery, rescoring, and report generation.
- **Automated Skill Optimization:** Developed a DeerFlow adapter for SkillOpt and an automated loop of candidate Skill injection, execution, evaluation, reflection, and Skill update, improving the Skill evaluation pass rate by 10 percentage points.
- **Proactive Flood Alert Assistant:** Enabled natural-language creation of alerts and one-time or recurring tasks; integrated API data retrieval, rule/LLM-based alert detection, record-level idempotency, on-duty personnel matching, unread escalation, and Feishu rich-media notifications.

HGTECH Co., Ltd. *AI Algorithm Intern* Jun. 2025 – Aug. 2025

Project: A RAG-based medical consultation system supporting symptom retrieval and disease knowledge matching.

- **RAG Pipeline:** Extracted medical textbook content with regular expressions into standardized JSON; built a LangChain and ChromaDB pipeline using BGE-M3 embeddings and CrossEncoder reranking, compared MMR and similarity retrieval, and implemented Top-20 recall followed by Top-5 reranking, achieving 88.27% symptom–disease matching accuracy.
- **Evaluation Framework:** Developed automated tests for verbatim clinical-presentation and generalized symptom retrieval, aggregated hit statistics, and generated timestamped visual reports.

PROJECTS

Deep Learning-Based Product Aesthetics Assessment Feb. 2026 – Apr. 2026

- **Model Design:** Formulated product aesthetics assessment as continuous regression; compared transfer-learned ResNet18, a lightweight CNN, and SqueezeNet2 in PyTorch under limited-data settings, achieving an MSE of 0.745 on the full dataset.
- **Personalized Aesthetic Prediction:** Built a multi-stream network from 21,600 ratings by 144 users, combining image features with 18-dimensional user attributes; used ablation studies to quantify the contribution of user information.

Collaborative Coding Platform Sept. 2025 – Nov. 2025

- **Real-Time Collaboration:** Used WebSocket and Yjs CRDT for real-time code editing, cursor synchronization, and state consistency; added Redis incremental persistence and Pub/Sub synchronization across instances.
- **Online Code Execution:** Designed a Ticket ID and Kafka-based asynchronous workflow for Python, C++, and JavaScript, covering compilation, test validation, result delivery, and status feedback.
- **Microservice Deployment:** Containerized and orchestrated services with Docker and Kubernetes, using Nginx Ingress for unified routing.

SKILLS

Languages: Mandarin Chinese (Putonghua Proficiency Test: Level 1-B); English (IELTS 7.5, CET-6 654).

Programming: Python, C++, C, Java; SQL querying and data processing; HTML, CSS, and JavaScript.

Agent Development: ReAct-style agent architectures, Tool Calling, Skills, RAG, and MCP.

LLMs & Deep Learning: Transformer fundamentals; model development and training with PyTorch and TensorFlow; SFT, LoRA, and QLoRA.

Backend & Engineering: Git, Linux, Docker, Kubernetes, Redis, and Kafka.